VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum- 590014, Karnataka.

LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

Vedanth CR(1BM23CS036)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

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B.M.S. College of Engineering

Bull Temple Road, Bangalore 560019

(Affiliated to Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Vedanth CR (1BM23CS036),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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8	09.04.26	Write a Scala program to print numbers from 1 to 100 using for loop.	33
9	09.04.26	Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.	34
10	16.04.26	Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).	35

Github Link: https://github.com/vedanthcr27/BDALAB

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

Commands with output:

Student database

1. Create a database “Student” with the following attributes Rollno, Age, ContactNo,

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.createCollection("Student");
{ ok: 1 }
```

Email-Id.

2. Insert appropriate values (at least 5)

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:1,Age:21,Cont:9876,email:"antara.de9@gmail.com"});
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ec7c840f42b4a05530") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:2,Age:22,Cont:9976,email:"anushka.de9@gmail.com"});

{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05531") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:3,Age:21,Cont:5576,email:"anubhav.de9@gmail.com"});

{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05532") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:4,Age:20,Cont:4476,email:"pani.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05533") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
```

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82f47c840f42b4a05534") }
```

3. View the data

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.find()
[
  {
    _id: ObjectId("660a82ec7c840f42b4a05530"),
    RollNo: 1,
    Age: 21,
    Cont: 9876,
    email: 'antara.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05531"),
    RollNo: 2,
    Age: 22,
    Cont: 9976,
    email: 'anushka.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05532"),
    RollNo: 3,
    Age: 21,
    Cont: 5576,
    email: 'anubhav.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05533"),
    RollNo: 4,
    Age: 20,
    Cont: 4476,
    email: 'pani.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82f47c840f42b4a05534"),
    RollNo: 10,
    Age: 23,
    Cont: 2276,
    email: 'rekha.de9@gmail.com'
  }
]
```

4. Write a query to update the Email-Id of a student with rollno 10.

```
  },
  {
    _id: ObjectId("660a83337c840f42b4a05535"),
    RollNo: 11,
    Age: 22,
    Name: 'ABC',
    Cont: 2276,
    email: 'rea.de9@gmail.com'
  }
]
```

5. Replace the student name from “ABC” to “FEM” of rollno 11.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.update({RollNo:11,Name:"ABC"},{$set:{Name:"FEM"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

6. Export the created table into local file system

```
C:\Users\BMSCECSE>mongoexport mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB  --collection=Student --out C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:15:56.383+0530    connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:15:56.497+0530    exported 7 records
```

7. Drop the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.drop();
true
```

8. Import a given csv dataset from local file system into mongodb collection.

```
C:\Users\BMSCECSE>mongoimport  mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB --collection=New_Student  --type json --file C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:17:35.523+0530    connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:17:35.640+0530    7 document(s) imported successfully. 0 document(s) failed to import.
```

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

Command with output using teminal:

```
test> use Employee
switched to db Employee
Employee> db
Employee
Employee> db.createCollection("Employee_Info")
{ ok: 1 }
Employee> show collections
Employee_Info
Employee> db.Employee_Info.insertMany([
...   {
...     Emp_Id: 121,
...     Emp_Name: "Amit",
...     Designation: "Software Engineer",
...     Date_of_Joining: ISODate("2019-06-15"),
...     Salary: 60000,
...     Dept_Name: "IT"
...   },
...   {
...     Emp_Id: 122,
...     Emp_Name: "Ravi",
...     Designation: "Senior Engineer",
...     Date_of_Joining: ISODate("2018-03-20"),
...     Salary: 75000,
...     Dept_Name: "R&D"
...   },
...   {
...     Emp_Id: 123,
...     Emp_Name: "Meera",
...     Designation: "Manager",
...     Date_of_Joining: ISODate("2016-01-10"),
...     Salary: 90000,
...     Dept_Name: "HR"
...   }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69899e77e7bd2cd77163b112'),
    '1': ObjectId('69899e77e7bd2cd77163b113'),
    '2': ObjectId('69899e77e7bd2cd77163b114')
  }
}
```

```
mongosh mongodb://127.0.0.

}
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: -1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b113'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: 1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
```

mongosh mongodb://127.0.0.

```
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: -1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b113'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  }
]
Employee> db.Employee_Info.find().sort({ Salary: 1 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b113'),
    Emp_Id: 122,
    Emp_Name: 'Ravi',
```

mongosh mongodb://127.0.0.

```
    Emp_Name: 'Ravi',
    Designation: 'Senior Engineer',
    Date_of_Joining: ISODate('2018-03-20T00:00:00.000Z'),
    Salary: 75000,
    Dept_Name: 'R&D'
  },
  {
    _id: ObjectId('69899e77e7bd2cd77163b114'),
    Emp_Id: 123,
    Emp_Name: 'Meera',
    Designation: 'Manager',
    Date_of_Joining: ISODate('2016-01-10T00:00:00.000Z'),
    Salary: 90000,
    Dept_Name: 'HR'
  }
]
Employee> db.Employee_Info.updateOne(
...    { Emp_Id: 121 },
...    { $set: { Projects: ["AI-Analytics", "Cloud-Migration", "BigData-Pipeline"] } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.updateOne(
...    { Emp_Id: 121 },
...    { $addToSet: { Projects: "ML-Deployment" } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science',
    Projects: [
```

mongosh mongodb://127.0.0.

```
}
Employee> db.Employee_Info.find({ Emp_Id: 121 })
[
  {
    _id: ObjectId('69899e77e7bd2cd77163b112'),
    Emp_Id: 121,
    Emp_Name: 'Amit Kumar',
    Designation: 'Software Engineer',
    Date_of_Joining: ISODate('2019-06-15T00:00:00.000Z'),
    Salary: 60000,
    Dept_Name: 'Data Science',
    Projects: [
      'AI-Analytics',
      'Cloud-Migration',
      'BigData-Pipeline',
      'ML-Deployment'
    ]
  }
]
Employee> db.Employee_Info.insertOne({
...   Emp_Id: 130,
...   Emp_Name: "TemporaryUser",
...   Designation: "Intern",
...   Date_of_Joining: ISODate("2024-01-01"),
...   Salary: 15000,
...   Dept_Name: "Training",
...   createdAt: new Date()
... })
...
{
  acknowledged: true,
  insertedId: ObjectId('69899ef7e7bd2cd77163b115')
}
Employee> db.Employee_Info.createIndex(
...   { createdAt: 1 },
...   { expireAfterSeconds: 15 }
... )
...
createdAt_1
Employee> db.Employee_Info.find({ Emp_Id: 130 })
[
  {
    _id: ObjectId('69899ef7e7bd2cd77163b115'),
    Emp_Id: 130,
    Emp_Name: 'TemporaryUser',
    Designation: 'Intern',
    Date_of_Joining: ISODate('2024-01-01T00:00:00.000Z'),
    Salary: 15000,
    Dept_Name: 'Training',
    createdAt: ISODate('2026-02-09T08:46:47.854Z')
  }
```

PROGRAM – 2

Perform the following DB operations using Cassandra Questions:

a) Create a keyspace by name Employee
b) Create a column family by name
- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch
d) Update Employee name and Department of Emp-Id 121
e) Sort the details of Employee records based on salary
f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
g) Update the altered table to add project names.
h) Create a TTL of 15 seconds to display the values of Employees.

Command with output

Employee Database

1. Create a keyspace by name Employee

```
cqlsh:library> CREATE KEYSPACE Employee WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh:library>
```

2. Create a column family by name Employee-Info with attributes Emp_Id Primary Key,

```
cqlsh:employee>
cqlsh:employee> CREATE TABLE Employee_Info (
            ...     Emp_Id int PRIMARY KEY,
            ...     Emp_Name text,
            ...     Designation text,
            ...     Date_of_Joining date,
            ...     Salary decimal,
            ...     Dept_Name text
            ... );
```

Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name

3. Insert the values into the table in batch

```
cqlsh:employee> BEGIN BATCH
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (101, 'John Doe', 'Manager', '2023-01-01', 50000, 'HR');
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (121, 'Jane Smith', 'Developer', '2023-02-01', 60000, 'IT');
            ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info SET Emp_Name = 'Jane Johnson', Dept_Name = 'Engineering' WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name   | designation | emp_name     | salary
--------+-----------------+-------------+-------------+--------------+--------
    121 |      2023-02-01 | Engineering |   Developer | Jane Johnson |  60000
    101 |      2023-01-01 |          HR |     Manager |     John Doe |  50000

(2 rows)
```

4. Update Employee name and Department of Emp-Id 121

5. Sort the details of Employee records based on salary

```
cqlsh:employee> paging off
Disabled Query paging.
cqlsh:employee> SELECT * FROM Employee_Info WHERE Emp_Id IN (121,101) ORDER BY Salary ALLOW FILTERING;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name   | projects
--------+--------+-----------------+-----------+-------------+------------+--------------------------
    121 |  60000 |      2023-02-01 |        IT |   Developer | Jane Smith |              {'ProjectC'}
    101 |  50000 |      2023-01-01 |        HR |     Manager |   John Doe | {'ProjectA', 'ProjectB'}

(2 rows)
```

6. Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.

7. Update the altered table to add project names.
8. Create a TTL of 15 seconds to display the values of Employees.

```
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name   | projects
--------+--------+-----------------+-----------+-------------+------------+--------------------------
    121 |  60000 |      2023-02-01 |        IT |   Developer | Jane Smith |              {'ProjectC'}
    101 |  50000 |      2023-01-01 |        HR |     Manager |   John Doe | {'ProjectA', 'ProjectB'}

(2 rows)
```

```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name) VALUES (102, 'Jane Smit
h', 'Developer', '2022-06-03', 60000, 'IT') USING TTL 15;
cqlsh:employee> select ttl(Emp_Name) from Employee_Info where Emp_id=102;

 ttl(emp_name)
---------------
            14

(1 rows)
```

PROGRAM – 3

Perform the following DB operations using Cassandra Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes
- Stud_Id Primary Key,
- Counter_value of type Counter,
- Stud_Name, Book-Name, Book-Id,
- Date_of_issue

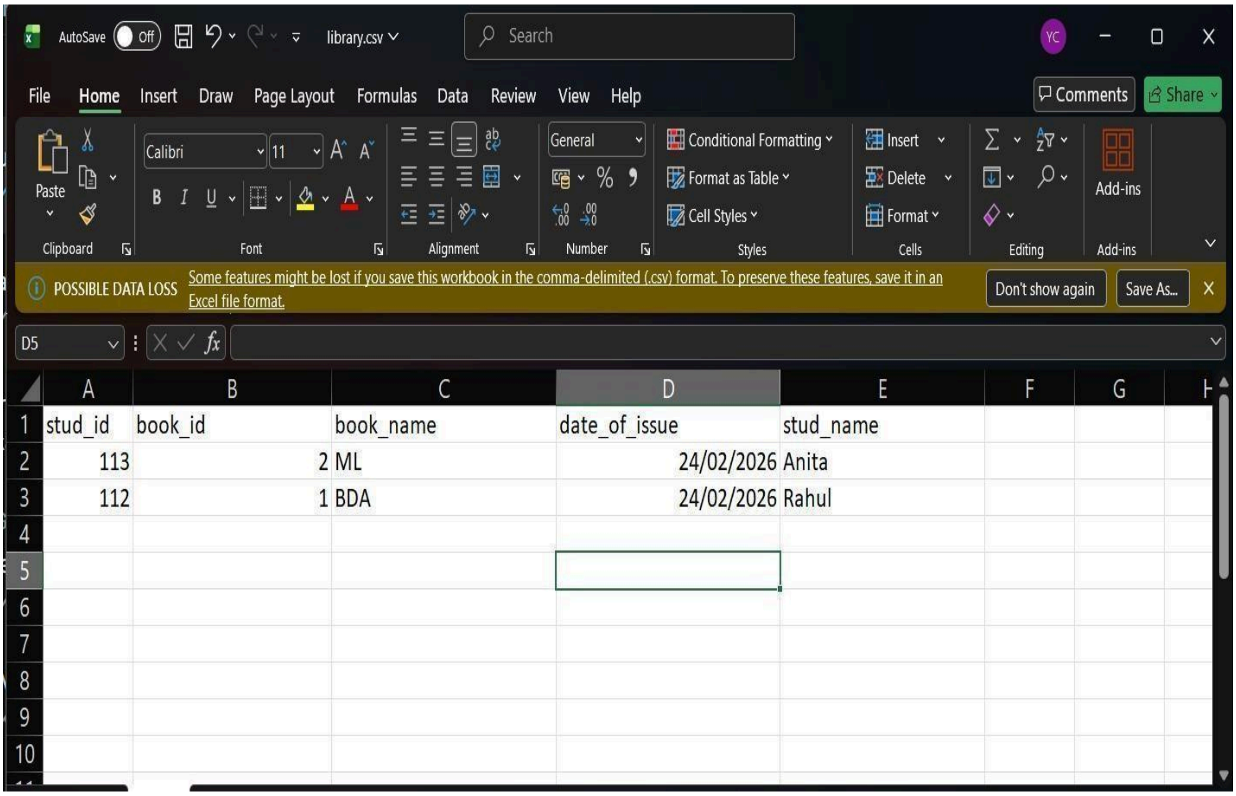
c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra

column family Command with output

```
Use HELP for help.
cqlsh> DESCRIBE KEYSPACES;

system       system_distributed  system_traces  system_virtual_schema
system_auth  system_schema       system_views

cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Book_Id int,
           ...     Date_of_issue date,
           ...     PRIMARY KEY (Stud_Id, Book_Id)
           ... );
cqlsh:library> CREATE TABLE Book_Count (
           ...     Stud_Id int,
           ...     Book_Name text,
           ...     Counter_value counter,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 1, '2026-02-24');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Anita', 'ML', 2, '2026-02-24');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Count
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```

```
          ... );
cqlsh:library> CREATE TABLE Book_Count (
          ...     Stud_Id int,
          ...     Book_Name text,
          ...     Counter_value counter,
          ...     PRIMARY KEY (Stud_Id, Book_Name)
          ... );
cqlsh:library> BEGIN BATCH
          ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
          ... VALUES (112, 'Rahul', 'BDA', 1, '2026-02-24');
          ...
          ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
          ... VALUES (113, 'Anita', 'ML', 2, '2026-02-24');
          ... APPLY BATCH;
cqlsh:library> UPDATE Book_Count
          ... SET Counter_value = Counter_value + 1
          ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Count
          ... SET Counter_value = Counter_value + 1
          ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |       2 |        ML |    2026-02-24 |     Anita
     112 |       1 |       BDA |    2026-02-24 |     Rahul

(2 rows)
cqlsh:library> SELECT * FROM Book_Count;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2
```

```
     112 |       BDA |             2

(1 rows)
cqlsh:library> UPDATE Book_Count
          ... SET Counter_value = Counter_value + 1
          ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT Counter_value
          ... FROM Book_Count
          ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 counter_value
---------------
             3

(1 rows)
cqlsh:library> exit

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker exec -it my-cassandra cqlsh -e "COPY Library.Library_Info
TO '/tmp/library.csv' WITH HEADER = true;"
Using 7 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:       7 rows/s; Avg. rate:       3 rows/s
2 rows exported to 1 files in 0.633 seconds.

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker cp my-cassandra:/tmp/library.csv library.csv
Successfully copied 2.05kB to C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin\library.csv

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker cp library.csv my-cassandra:/tmp/library.csv
Successfully copied 2.05kB to my-cassandra:/tmp/library.csv

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker exec -it my-cassandra cqlsh -e "COPY Library.Library_Info
FROM '/tmp/library.csv' WITH HEADER = true;"
Using 7 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:       4 rows/s; Avg. rate:       5 rows/s
2 rows imported from 1 files in 0.379 seconds (0 skipped).
```


AutoSave Off
library.csv
Search
File Home Insert Draw Page Layout Formulas Data Review View Help
Comments
Share
Paste
Calibri
11
General
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Add-ins
Clipboard Font Alignment Number Styles Cells Editing Add-ins
POSSIBLE DATA LOSS
Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format.
Don't show again
Save As...
D5
stud_id book_id book_name date_of_issue stud_name
113 2 ML 24/02/2026 Anita
112 1 BDA 24/02/2026 Rahul

PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

Command with output

1. mkdir
2. ls

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir /bda_hadoop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 1 items
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 14:37 /bda_hadoop
```

3.put

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -put /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /bda_hadoop
Found 1 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file.txt
Hello!!!
```

4. copyFromLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

5. get

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -get /bda_hadoop/file.txt /home/hadoop/Desktop/downloaded_file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -getmerge /bda_hadoop/file.txt /bda_hadoop/file_cp_local.txt /home/hadoop/Desktop/downloaded_file.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -getfacl /bda_hadoop/
# file: /bda_hadoop
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

6. copyToLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyToLocal /bda_hadoop/file.txt /home/hadoop/Desktop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: `/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

7. cat

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

8.mv

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
```

9.cp

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: `/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

PROGRAM – 5

Implement Wordcount program on Hadoop framework

Driver Code

```
import java.io.IOException;
import
org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import
org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import
org.apache.hadoop.util.ToolRunner;
public class WCDriver extends Configured implements Tool {

  public int run(String[] args) throws IOException {

    if (args.length < 2) {
      System.out.println("Please give valid
      inputs"); return -1;
    }

    JobConf conf = new JobConf(WCDriver.class);
    conf.setJobName("WordCount");

    FileInputFormat.setInputPaths(conf, new Path(args[0]));
    FileOutputFormat.setOutputPath(conf, new
    Path(args[1]));

    conf.setMapperClass(WCMapper.class);
    conf.setReducerClass(WCReducer.class
    );

    conf.setMapOutputKeyClass(Text.class);
    conf.setMapOutputValueClass(IntWritable.clas
    s);

    conf.setOutputKeyClass(Text.class);
    conf.setOutputValueClass(IntWritable.clas
    s);

    JobClient.runJob(conf)
    ; return 0;
  }

  // Main Method
  public static void main(String[] args) throws Exception {
    int exitCode = ToolRunner.run(new WCDriver(),
    args); System.out.println("Job Exit Code: " +
    exitCode);
  }
}
```

Mapper Code

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import
org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import
```

```
org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import
org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
```

```
public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
            throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+"))
        {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}
```

Reducer Code

```
  import
java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import
org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable>
                    values, OutputCollector<Text, IntWritable>
                    output, Reporter reporter) throws
                    IOException {
        int count = 0;

        // Counting the frequency of each
        word while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}
```

```
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ mkdir WordCount
cd WordCount
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCMapper.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCReducer.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCDriver.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ export HADOOP_CLASSPATH=$(hadoop classpath)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ javac -classpath $HADOOP_CLASSPATH -d . WCMapper.java WCReducer.java WCDriver.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ jar -cvf wordcount.jar *.class
added manifest
adding: WCDriver.class(in = 1397) (out= 689)(deflated 50%)
adding: WCMapper.class(in = 1857) (out= 757)(deflated 59%)
adding: WCReducer.class(in = 1529) (out= 605)(deflated 60%)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ cd /usr/local/hadoop/sbin
start-all.sh
jps
bash: cd: /usr/local/hadoop/sbin: No such file or directory
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-600-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
9152 Jps
8274 SecondaryNameNode
7977 DataNode
7882 NameNode
8570 ResourceManager
8732 NodeManager
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ mkdir input
nano input/sample.txt
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -mkdir /inputbda
hadoop fs -put input/sample.txt /inputbda
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop jar wordcount.jar WCDriver /inputbda/sample.txt /outputbda
2026-04-24 14:17:12,433 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-24 14:17:12,470 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-24 14:17:12,470 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-24 14:17:12,477 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-24 14:17:12,533 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-24 14:17:12,579 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-24 14:17:12,607 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1387788888_0001
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-24 14:17:12,720 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-24 14:17:12,720 INFO mapreduce.Job: Running job: job_local1387788888_0001
2026-04-24 14:17:12,721 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-24 14:17:12,722 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,762 INFO mapred.LocalJobRunner: Waiting for map tasks
```


```
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -mkdir /inputbda
hadoop fs -put input/sample.txt /inputbda
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop jar wordcount.jar WCDriver /inputbda/sample.txt /outputbda
2026-04-24 14:17:12,433 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-24 14:17:12,470 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-24 14:17:12,470 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-24 14:17:12,477 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-24 14:17:12,533 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-24 14:17:12,579 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-24 14:17:12,607 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1387788888_0001
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-24 14:17:12,720 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-24 14:17:12,720 INFO mapreduce.Job: Running job: job_local1387788888_0001
2026-04-24 14:17:12,721 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-24 14:17:12,722 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,762 INFO mapred.LocalJobRunner: Waiting for map tasks
```


```
2026-04-24 14:17:12,764 INFO mapred.LocalJobRunner: Starting task: attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,775 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,775 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,780 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-04-24 14:17:12,784 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/inputbda/sample.txt:0+89
2026-04-24 14:17:12,793 INFO mapred.MapTask: numReduceTasks: 1
2026-04-24 14:17:12,821 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-04-24 14:17:12,821 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-04-24 14:17:12,821 INFO mapred.MapTask: soft limit at 83886080
2026-04-24 14:17:12,821 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-04-24 14:17:12,821 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-04-24 14:17:12,823 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-04-24 14:17:12,884 INFO mapred.LocalJobRunner:
2026-04-24 14:17:12,884 INFO mapred.MapTask: Starting flush of map output
2026-04-24 14:17:12,884 INFO mapred.MapTask: Spilling map output
2026-04-24 14:17:12,884 INFO mapred.MapTask: bufstart = 0; bufend = 169; bufvoid = 104857600
2026-04-24 14:17:12,884 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214320(104857280); length = 77/6553600
2026-04-24 14:17:12,887 INFO mapred.MapTask: Finished spill 0
2026-04-24 14:17:12,892 INFO mapred.Task: Task:attempt_local1387788888_0001_m_000000_0 is done. And is in the process of committing
2026-04-24 14:17:12,894 INFO mapred.LocalJobRunner: hdfs://localhost:9000/inputbda/sample.txt:0+89
2026-04-24 14:17:12,894 INFO mapred.Task: Task 'attempt_local1387788888_0001_m_000000_0' done.
2026-04-24 14:17:12,897 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=2920
                FILE: Number of bytes written=644387
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=89
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=5
```

```
                Map input records=5
                Map output records=20
                Map output bytes=169
                Map output materialized bytes=215
                Input split bytes=93
                Combine input records=0
                Spilled Records=20
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=526385152
        File Input Format Counters
                Bytes Read=89
2026-04-24 14:17:12,897 INFO mapred.LocalJobRunner: Finishing task: attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,897 INFO mapred.LocalJobRunner: map task executor complete.
2026-04-24 14:17:12,899 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-04-24 14:17:12,899 INFO mapred.LocalJobRunner: Starting task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:12,902 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,902 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,902 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-04-24 14:17:12,904 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@53ac8b1c
2026-04-24 14:17:12,904 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-24 14:17:12,911 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=5830921216, maxSingleShuffleLimit=1457730304, mergeThreshold=3848408064, ioSortFactor=10, memToMemMergeOutputsThreshold=
2026-04-24 14:17:12,912 INFO reduce.EventFetcher: attempt_local1387788888_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-04-24 14:17:12,925 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1387788888_0001_m_000000_0 decomp: 211 len: 215 to MEMORY
2026-04-24 14:17:12,926 INFO reduce.InMemoryMapOutput: Read 211 bytes from map-output for attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,927 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 211, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->211
2026-04-24 14:17:12,927 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-04-24 14:17:12,928 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,928 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-04-24 14:17:12,930 INFO mapred.Merger: Merging 1 sorted segments
2026-04-24 14:17:12,930 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 205 bytes
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merged 1 segments, 211 bytes to disk to satisfy reduce memory limit
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merging 1 files, 215 bytes from disk
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-04-24 14:17:12,931 INFO mapred.Merger: Merging 1 sorted segments
2026-04-24 14:17:12,931 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 205 bytes
2026-04-24 14:17:12,931 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,991 INFO mapred.Task: Task:attempt_local1387788888_0001_r_000000_0 is done. And is in the process of committing
2026-04-24 14:17:12,992 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,992 INFO mapred.Task: Task attempt_local1387788888_0001_r_000000_0 is allowed to commit now
2026-04-24 14:17:13,007 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1387788888_0001_r_000000_0' to hdfs://localhost:9000/outputbda
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce > reduce
2026-04-24 14:17:13,008 INFO mapred.Task: Task 'attempt_local1387788888_0001_r_000000_0' done.
2026-04-24 14:17:13,008 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_r_000000_0: Counters: 30
        File System Counters
                FILE: Number of bytes read=3382
                FILE: Number of bytes written=644602
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=89
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
```

```
2026-04-24 14:17:12,992 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,992 INFO mapred.Task: Task attempt_local1387788888_0001_r_000000_0 is allowed to commit now
2026-04-24 14:17:13,007 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1387788888_0001_r_000000_0' to hdfs://localhost:9000/outputbda
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce > reduce
2026-04-24 14:17:13,008 INFO mapred.Task: Task 'attempt_local1387788888_0001_r_000000_0' done.
2026-04-24 14:17:13,008 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_r_000000_0: Counters: 30
        File System Counters
                FILE: Number of bytes read=3382
                FILE: Number of bytes written=644602
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=89
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
Ubuntu Software uce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=10
                Reduce shuffle bytes=215
                Reduce input records=20
                Reduce output records=10
                Spilled Records=20
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=526385152
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=69
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: Finishing task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-04-24 14:17:13,723 INFO mapreduce.Job: Job job_local1387788888_0001 running in uber mode : false
2026-04-24 14:17:13,723 INFO mapreduce.Job:  map 100% reduce 100%
2026-04-24 14:17:13,724 INFO mapreduce.Job: Job job_local1387788888_0001 completed successfully
2026-04-24 14:17:13,728 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=6302
                FILE: Number of bytes written=1288989
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=178
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=15
```

```
                FILE: Number of bytes written=1288989
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=178
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=5
                Map output records=20
                Map output bytes=169
                Map output materialized bytes=215
                Input split bytes=93
                Combine input records=0
                Combine output records=0
                Reduce input groups=10
                Reduce shuffle bytes=215
                Reduce input records=20
                Reduce output records=10
                Spilled Records=40
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1052770304
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=89
        File Output Format Counters
                Bytes Written=69
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -ls /outputbda
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2026-04-24 14:17 /outputbda/_SUCCESS
-rw-r--r--   1 hadoop supergroup         69 2026-04-24 14:17 /outputbda/part-00000
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -cat /outputbda/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$
```

PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncd
c/all
a) Create a MapReduce program to find average temperature for each year from NCDC data set.
b) find the mean max temperature for every month.

Driver Code

```
package temp;

import
org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {
  public static void main(String[] args) throws Exception {
    if (args.length != 2) {
      System.err.println("Please enter both input and output
      parameters."); System.exit(-1);
    }

    // Creating a configuration and job
    instance Configuration conf = new
    Configuration();
    Job job = Job.getInstance(conf, "Average

    Calculation"); job.setJarByClass(AverageDriver.class);

    // Input and output paths
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new
    Path(args[1]));

    // Setting mapper and reducer classes
    job.setMapperClass(AverageMapper.class)
    ;
    job.setReducerClass(AverageReducer.clas
    s);

    // Output key and value types
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.clas
    s);

    // Submitting the job and waiting for it to complete
```

```
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

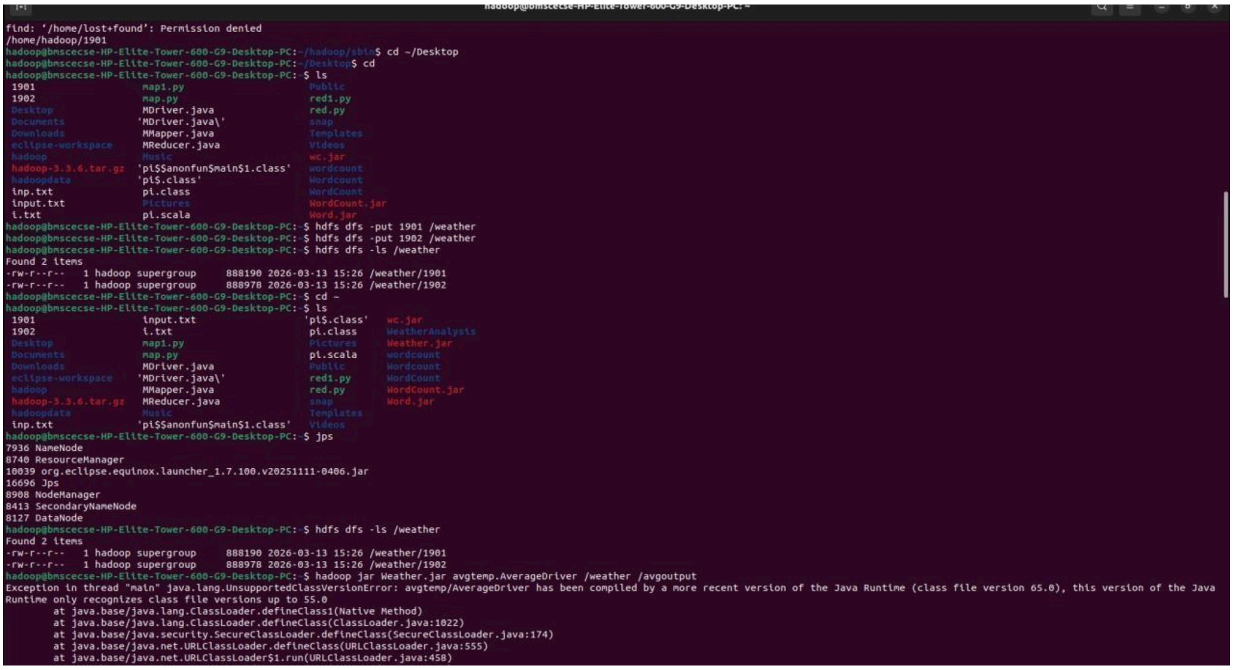
Mapper Code

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import
org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable>

  { public static final int MISSING = 9999;

  @Override
  public void map(LongWritable key, Text value, Context
      context) throws IOException, InterruptedException {

    String line = value.toString();

    // Extract year from fixed position
    String year = line.substring(15,
    19); int temperature;

    // Determine if there's a '+'
    sign if (line.charAt(87) ==
    '+') {
      temperature = Integer.parseInt(line.substring(88, 92));
    } else {
      temperature = Integer.parseInt(line.substring(87, 92));
    }

    // Quality check character
    String quality = line.substring(92, 93);

    // Only emit if data is valid
    if (temperature != MISSING && quality.matches("[01459]")) {
      context.write(new Text(year), new IntWritable(temperature));
    }
  }
}
```

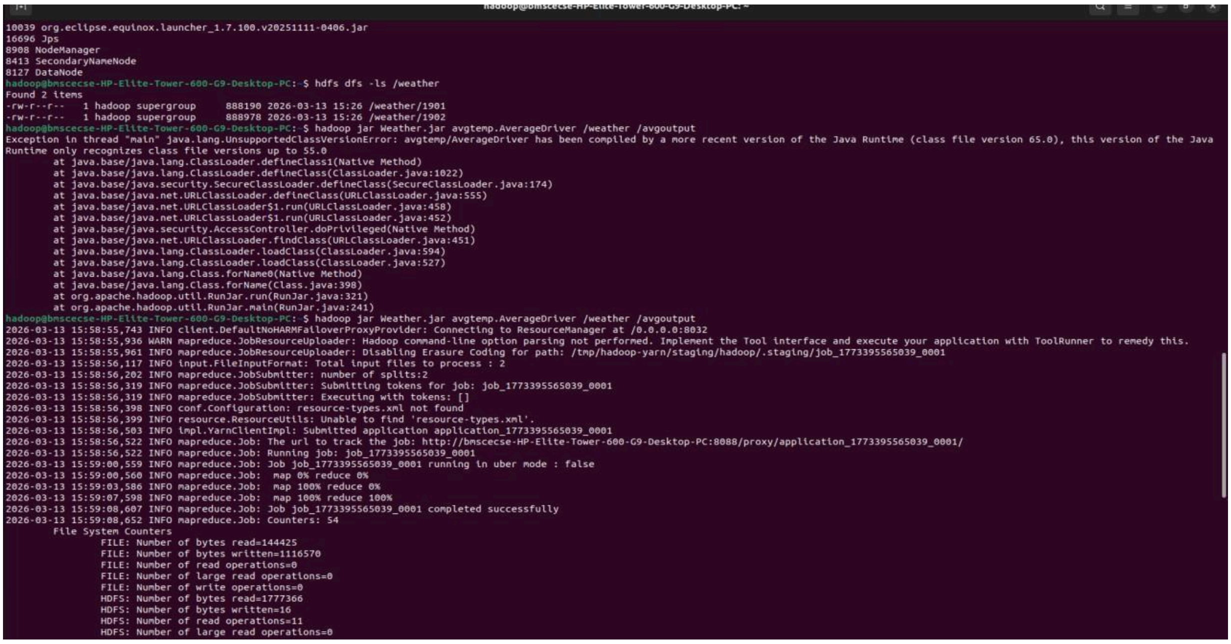
Reducer Code

```
package temp;

import java.io.IOException;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
  @Override
  public void reduce(Text key, Iterable<IntWritable> values,
          Context context) throws IOException, InterruptedException {

    int sumTemp =
    0; int count = 0;

    for (IntWritable value : values)
      { sumTemp += value.get();
      count++;
    }

    if (count > 0) {
      int average = sumTemp / count;
      context.write(key, new IntWritable(average));
```

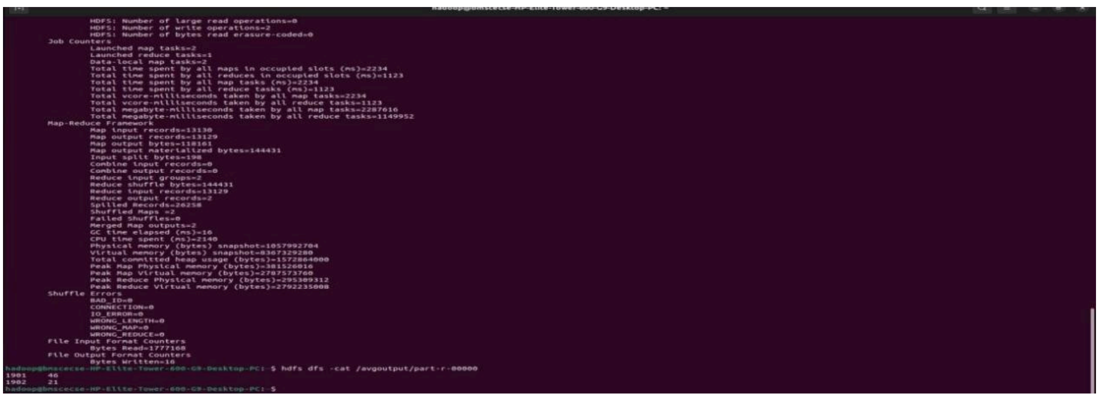
}
}
}



find: '/home/lost+found': Permission denied
/home/hadoop/1901
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ cd ~/Desktop
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/Desktop$ cd
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ ls
1901 map1.py Public
1902 map.py red1.py
Desktop MDriver.java red.py
Documents 'MDriver.java\' snap
Downloads MMapper.java Templates
eclipse-workspace MReducer.java Videos
hadoop Music wc.jar
hadoop-3.3.6.tar.gz 'pi$$anonfun$main$1.class' wordcount
hadoopdata 'pi$.class' Wordcount
inp.txt pi.class WordCount
input.txt Pictures WordCount.jar
i.txt pi.scala Word.jar
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -put 1901 /weather
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -put 1902 /weather
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -ls /weather
Found 2 items
-rw-r--r-- 1 hadoop supergroup 888190 2026-03-13 15:26 /weather/1901
-rw-r--r-- 1 hadoop supergroup 888978 2026-03-13 15:26 /weather/1902
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ cd ~
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ ls
1901 input.txt 'pi$.class' wc.jar
1902 i.txt pi.class WeatherAnalysis
Desktop map1.py Pictures Weather.jar
Documents map.py pi.scala wordcount
Downloads MDriver.java Public Wordcount
eclipse-workspace 'MDriver.java\' red1.py WordCount
hadoop MMapper.java red.py WordCount.jar
hadoop-3.3.6.tar.gz MReducer.java snap Word.jar
hadoopdata Music Templates
inp.txt 'pi$$anonfun$main$1.class' Videos
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ jps
7936 NameNode
8740 ResourceManager
10039 org.eclipse.equinox.launcher_1.7.100.v20251111-0406.jar
16696 Jps
8908 NodeManager
8413 SecondaryNameNode
8127 DataNode
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -ls /weather
Found 2 items
-rw-r--r-- 1 hadoop supergroup 888190 2026-03-13 15:26 /weather/1901
-rw-r--r-- 1 hadoop supergroup 888978 2026-03-13 15:26 /weather/1902
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
Exception in thread "main" java.lang.UnsupportedClassVersionError: avgtemp/AverageDriver has been compiled by a more recent version of the Java Runtime (class file version 65.0), this version of the Java Runtime only recognizes class file versions up to 55.0
at java.base/java.lang.ClassLoader.defineClass1(Native Method)
at java.base/java.lang.ClassLoader.defineClass(ClassLoader.java:1022)
at java.base/java.security.SecureClassLoader.defineClass(SecureClassLoader.java:174)
at java.base/java.net.URLClassLoader.defineClass(URLClassLoader.java:555)
at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:458)



10039 org.eclipse.equinox.launcher_1.7.100.v20251111-0406.jar
16696 Jps
8908 NodeManager
8413 SecondaryNameNode
8127 DataNode
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -ls /weather
Found 2 items
-rw-r--r-- 1 hadoop supergroup 888190 2026-03-13 15:26 /weather/1901
-rw-r--r-- 1 hadoop supergroup 888978 2026-03-13 15:26 /weather/1902
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
Exception in thread "main" java.lang.UnsupportedClassVersionError: avgtemp/AverageDriver has been compiled by a more recent version of the Java Runtime (class file version 65.0), this version of the Java Runtime only recognizes class file versions up to 55.0
at java.base/java.lang.ClassLoader.defineClass1(Native Method)
at java.base/java.lang.ClassLoader.defineClass(ClassLoader.java:1022)
at java.base/java.security.SecureClassLoader.defineClass(SecureClassLoader.java:174)
at java.base/java.net.URLClassLoader.defineClass(URLClassLoader.java:555)
at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:458)
at java.base/java.net.URLClassLoader$1.run(URLClassLoader.java:452)
at java.base/java.security.AccessController.doPrivileged(Native Method)
at java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:451)
at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:594)
at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:527)
at java.base/java.lang.Class.forName0(Native Method)
at java.base/java.lang.Class.forName(Class.java:398)
at org.apache.hadoop.util.RunJar.run(RunJar.java:321)
at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar avgtemp.AverageDriver /weather /avgoutput
2026-03-13 15:58:55,743 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-13 15:58:55,936 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-13 15:58:55,961 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1773395565039_0001
2026-03-13 15:58:56,117 INFO input.FileInputFormat: Total input files to process : 2
2026-03-13 15:58:56,202 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-13 15:58:56,319 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773395565039_0001
2026-03-13 15:58:56,319 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-13 15:58:56,398 INFO conf.Configuration: resource-types.xml not found
2026-03-13 15:58:56,399 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-13 15:58:56,503 INFO impl.YarnClientImpl: Submitted application application_1773395565039_0001
2026-03-13 15:58:56,522 INFO mapreduce.Job: The url to track the job: http://bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:8088/proxy/application_1773395565039_0001/
2026-03-13 15:58:56,522 INFO mapreduce.Job: Running job: job_1773395565039_0001
2026-03-13 15:59:00,559 INFO mapreduce.Job: Job job_1773395565039_0001 running in uber mode : false
2026-03-13 15:59:00,560 INFO mapreduce.Job: map 0% reduce 0%
2026-03-13 15:59:03,586 INFO mapreduce.Job: map 100% reduce 0%
2026-03-13 15:59:07,598 INFO mapreduce.Job: map 100% reduce 100%
2026-03-13 15:59:08,607 INFO mapreduce.Job: Job job_1773395565039_0001 completed successfully
2026-03-13 15:59:08,652 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=144425
FILE: Number of bytes written=1116570
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=1777366
HDFS: Number of bytes written=16
HDFS: Number of read operations=11
HDFS: Number of large read operations=0



HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0
Job Counters
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=2234
Total time spent by all reduces in occupied slots (ms)=1123
Total time spent by all map tasks (ms)=2234
Total time spent by all reduce tasks (ms)=1123
Total vcore-milliseconds taken by all map tasks=2234
Total vcore-milliseconds taken by all reduce tasks=1123
Total megabyte-milliseconds taken by all map tasks=2287616
Total megabyte-milliseconds taken by all reduce tasks=1149952
Map-Reduce Framework
Map input records=13130
Map output records=13129
Map output bytes=118161
Map output materialized bytes=144431
Input split bytes=198
Combine input records=0
Combine output records=0
Reduce input groups=2
Reduce shuffle bytes=144431
Reduce input records=13129
Reduce output records=2
Spilled Records=26258
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=16
CPU time spent (ms)=2140
Physical memory (bytes) snapshot=1057992704
Virtual memory (bytes) snapshot=8367329280
Total committed heap usage (bytes)=1572864000
Peak Map Physical memory (bytes)=381526016
Peak Map Virtual memory (bytes)=2787573760
Peak Reduce Physical memory (bytes)=295309312
Peak Reduce Virtual memory (bytes)=2792235008
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=1777168
File Output Format Counters
Bytes Written=16
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -cat /avgoutput/part-r-00000
1901 46
1902 21
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$

Driver Code

```
package meanmax;

import
org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {
  public static void main(String[] args) throws Exception {
    if (args.length != 2) {
      System.err.println("Please enter both input and output
      parameters."); System.exit(-1);
    }

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Mean and Max

    Temperature"); job.setJarByClass(MeanMaxDriver.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new
    Path(args[1]));

    job.setMapperClass(MeanMaxMapper.class
    );
    job.setReducerClass(MeanMaxReducer.clas
    s);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.clas
    s);

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

Mapper Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import
org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable>

  { public static final int MISSING = 9999;

  @Override
  public void map(LongWritable key, Text value, Context
      context) throws IOException, InterruptedException {

    String line = value.toString();
```

```
// Extract month from positions 19-20
String month = line.substring(19, 21);
int temperature;
```

```
        // Extract temperature considering optional
        '+' if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check
        String quality = line.substring(92, 93);

        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(month), new
            IntWritable(temperature));
        }
    }
}
```

Reducer Code

```
package meanmax;

import java.io.IOException;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {
    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp =
        0; int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values)
            { int temp = value.get();
            sumTemp += temp;
            count++;

            if (temp > maxTemp)
                { maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
            context.write(key, new Text(result));
        }
    }
}
```

```
1901	46
1902	21
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -rm -r /meanmaxoutput
rm: `/meanmaxoutput': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hadoop jar Weather.jar meanmax.MeanMaxDriver /weather /meanmaxoutput
2026-03-13 16:05:30,143 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-13 16:05:30,276 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-13 16:05:30,305 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1773395565039_0002
2026-03-13 16:05:30,453 INFO input.FileInputFormat: Total input files to process : 2
2026-03-13 16:05:30,529 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-13 16:05:30,612 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1773395565039_0002
2026-03-13 16:05:30,612 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-13 16:05:30,682 INFO conf.Configuration: resource-types.xml not found
2026-03-13 16:05:30,682 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-13 16:05:30,709 INFO impl.YarnClientImpl: Submitted application application_1773395565039_0002
2026-03-13 16:05:30,725 INFO mapreduce.Job: The url to track the job: http://bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:8088/proxy/application_1773395565039_0002/
2026-03-13 16:05:30,726 INFO mapreduce.Job: Running job: job_1773395565039_0002
2026-03-13 16:05:34,760 INFO mapreduce.Job: Job job_1773395565039_0002 running in uber mode : false
2026-03-13 16:05:34,761 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-13 16:05:37,786 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-13 16:05:41,800 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-13 16:05:41,806 INFO mapreduce.Job: Job job_1773395565039_0002 completed successfully
2026-03-13 16:05:41,847 INFO mapreduce.Job: Counters: 54
	File System Counters
		FILE: Number of bytes read=118167
		FILE: Number of bytes written=1064069
		FILE: Number of read operations=0
		FILE: Number of large read operations=0
		FILE: Number of write operations=0
		HDFS: Number of bytes read=1777366
		HDFS: Number of bytes written=72
		HDFS: Number of read operations=11
		HDFS: Number of large read operations=0
		HDFS: Number of write operations=2
		HDFS: Number of bytes read erasure-coded=0
	Job Counters
		Launched map tasks=2
		Launched reduce tasks=1
		Data-local map tasks=2
		Total time spent by all maps in occupied slots (ms)=2213
		Total time spent by all reduces in occupied slots (ms)=1047
		Total time spent by all map tasks (ms)=2213
		Total time spent by all reduce tasks (ms)=1047
		Total vcore-milliseconds taken by all map tasks=2213
		Total vcore-milliseconds taken by all reduce tasks=1047
		Total megabyte-milliseconds taken by all map tasks=2266112
		Total megabyte-milliseconds taken by all reduce tasks=1072128
	Map-Reduce Framework
		Map input records=13130
		Map output records=13129
		Map output bytes=91903
		Map output materialized bytes=118173
		Input split bytes=198
		Combine input records=0
		Combine output records=0
```

```
		Total time spent by all reduce tasks (ms)=1047
		Total vcore-milliseconds taken by all map tasks=2213
		Total vcore-milliseconds taken by all reduce tasks=1047
		Total megabyte-milliseconds taken by all map tasks=2266112
		Total megabyte-milliseconds taken by all reduce tasks=1072128
	Map-Reduce Framework
		Map input records=13130
		Map output records=13129
		Map output bytes=91903
		Map output materialized bytes=118173
		Input split bytes=198
		Combine input records=0
		Combine output records=0
		Reduce input groups=12
		Reduce shuffle bytes=118173
		Reduce input records=13129
		Reduce output records=12
		Spilled Records=26258
		Shuffled Maps =2
		Failed Shuffles=0
		Merged Map outputs=2
		GC time elapsed (ms)=15
		CPU time spent (ms)=2030
		Physical memory (bytes) snapshot=1037684736
		Virtual memory (bytes) snapshot=8376963072
		Total committed heap usage (bytes)=1572864000
		Peak Map Physical memory (bytes)=381546496
		Peak Map Virtual memory (bytes)=2792185856
		Peak Reduce Physical memory (bytes)=281702400
		Peak Reduce Virtual memory (bytes)=2799857664
	Shuffle Errors
		BAD_ID=0
		CONNECTION=0
		IO_ERROR=0
		WRONG_LENGTH=0
		WRONG_MAP=0
		WRONG_REDUCE=0
	File Input Format Counters
		Bytes Read=1777168
	File Output Format Counters
		Bytes Written=72
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ hdfs dfs -cat /meanmaxoutput/part-r-00000
01	4
02	1
03	6
04	34
05	89
06	143
07	182
08	172
09	123
10	73
11	21
12	3
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$
```

PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

Driver Code (TopNDriver.java)

```
package samples.topn;

import
org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

  public static void main(String[] args) throws Exception {
    if (args.length != 3) {
      System.err.println("Usage: TopNDriver <in> <temp-out>
      <final-out>"); System.exit(2);
    }

    Configuration conf = new Configuration();

    // === Job 1: Word Count ===
    Job wcJob = Job.getInstance(conf, "word count");
    wcJob.setJarByClass(TopNDriver.class);
    wcJob.setMapperClass(WordCountMapper.class);
    wcJob.setCombinerClass(WordCountReducer.clas
    s);
    wcJob.setReducerClass(WordCountReducer.class)
    ; wcJob.setOutputKeyClass(Text.class);
    wcJob.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(wcJob, new Path(args[0]));
    Path tempDir = new Path(args[1]);
    FileOutputFormat.setOutputPath(wcJob, tempDir);

    if (!wcJob.waitForCompletion(true))
      { System.exit(1);
    }

    // === Job 2: Top N ===
    Job topJob = Job.getInstance(conf, "top 10
    words"); topJob.setJarByClass(TopNDriver.class);
    topJob.setMapperClass(TopNMapper.class);
    topJob.setReducerClass(TopNReducer.class);
    topJob.setMapOutputKeyClass(IntWritable.class);
    topJob.setMapOutputValueClass(Text.class);
    topJob.setOutputKeyClass(Text.class);
    topJob.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(topJob, tempDir);
    FileOutputFormat.setOutputPath(topJob, new
    Path(args[2]));
```

```
        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import
java.util.StringTokenizer;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper
  extends Mapper<Object, Text, Text, IntWritable> {

  private final static IntWritable ONE = new
  IntWritable(1); private Text word = new Text();
  // characters to normalize into spaces
  private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\""]";

  @Override
  protected void map(Object key, Text value, Context
    context) throws IOException, InterruptedException {

    // clean & tokenize
    String clean = value.toString()
                .toLowerCase()
                .replaceAll(tokens, " ");
    StringTokenizer itr = new
    StringTokenizer(clean); while
    (itr.hasMoreTokens()) {
      word.set(itr.nextToken().trim());
      context.write(word, ONE);
    }
  }
}
```

Mapper Code (TopNMapper.java)

```
package samples.topn;

import

java.io.IOException;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper
  extends Mapper<Object, Text, IntWritable, Text> {

  private IntWritable count = new
  IntWritable(); private Text word = new
  Text();

  @Override
  protected void map(Object key, Text value, Context
    context) throws IOException, InterruptedException {

    // input line: word \t count
    String[] parts =
    value.toString().split("\\t"); if (parts.length
    == 2) {
      word.set(parts[0]);
      count.set(Integer.parseInt(parts[1])
      );
      // emit count → word, so Hadoop sorts by count
      context.write(count, word);
    }
  }
```

}

Reducer Code (WordCountReducer.java)

package samples.topn;

import

java.io.IOException;

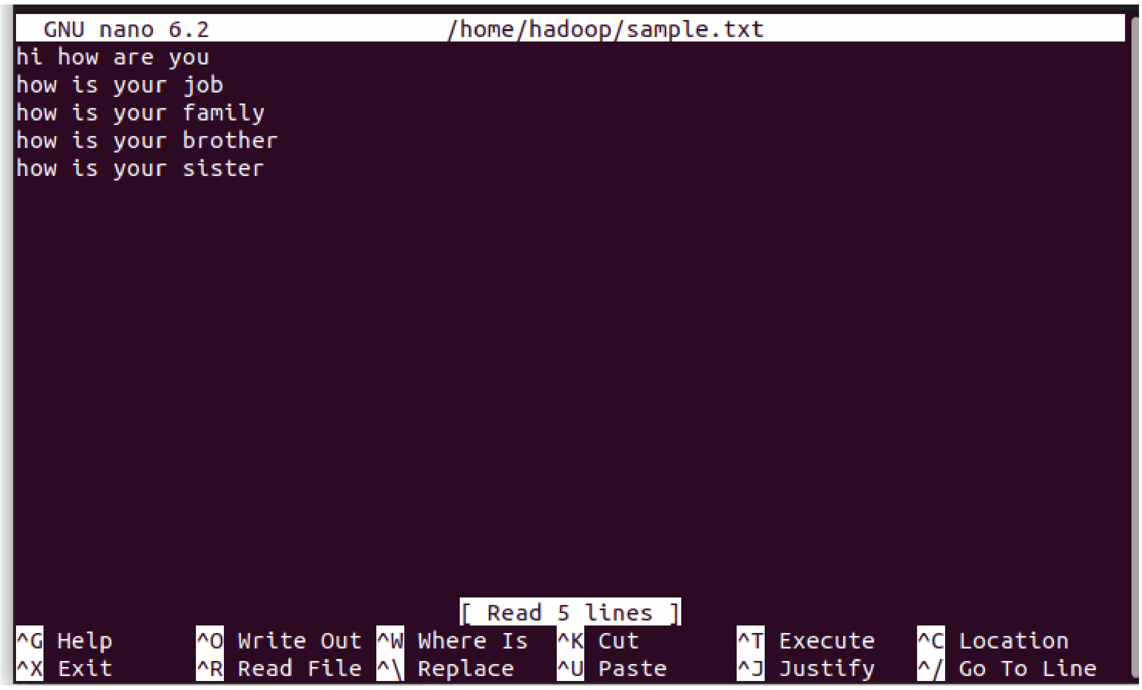
```
import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context
        context) throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values)
            { sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

Reducer Code (TopNReducer.java)

```
package samples.topn;

import
java.io.IOException;
import java.util.ArrayList;
import
java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer
    extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap
    =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context
        context) throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new
        ArrayList<>()); for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
        throws IOException, InterruptedException
        {

        // collect top 10 word→count pairs
        List<WordCount> topList = new
        ArrayList<>(); int seen = 0;
        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet())
            { int cnt = entry.getKey();
            for (String w : entry.getValue()) {
                topList.add(new WordCount(w,
                cnt)); seen++;
                if (seen == 10) break;
```

```
}
if (seen == 10) break;
```

```
        }

        // sort these 10 entries alphabetically by word
        Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

        // emit final top 10 in alphabetical
        order for (WordCount wc : topList) {
            context.write(new Text(wc.word), new IntWritable(wc.count));
        }
    }

    // helper class
    private static class WordCount
        { String word;
        int count;
        WordCount(String w, int c) { word = w; count = c; }
    }
}
```



```
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ ./start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-600-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ jps
31057 Jps
30787 NodeManager
30149 SecondaryNameNode
29706 NameNode
29902 DataNode
30431 ResourceManager
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/hadoop/sbin$ cd ~/topn
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ javac -classpath `hadoop classpath` -d . TopN.java TopNMapper.java TopNReducer.java TopNCombiner.java utils/MiscUtils.java
Note: utils/MiscUtils.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ jar -cvf TopN.jar *.class utils/*.class
added manifest
adding: TopN.class(in = 1388) (out= 764)(deflated 44%)
adding: TopNCombiner.class(in = 1587) (out= 662)(deflated 58%)
adding: TopNMapper.class(in = 1953) (out= 883)(deflated 54%)
adding: TopNReducer.class(in = 2532) (out= 1003)(deflated 60%)
adding: utils/MiscUtils$1.class(in = 944) (out= 494)(deflated 47%)
adding: utils/MiscUtils.class(in = 1231) (out= 693)(deflated 43%)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ nano ~/sample.txt
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -mkdir /topninput
mkdir: `/topninput': File exists
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -put ~/sample.txt /topninput
put: `/topninput/sample.txt': File exists
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -ls /topninput
Found 1 items
-rw-r--r--   1 hadoop supergroup         89 2026-04-24 14:58 /topninput/sample.txt
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop jar TopN.jar TopN /topninput/sample.txt /topnoutput
2026-04-24 15:04:22,185 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
Exception in thread "main" org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/topnoutput already exists
        at org.apache.hadoop.mapreduce.lib.output.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:164)
        at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:277)
        at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:143)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1678)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1675)
        at java.base/java.security.AccessController.doPrivileged(Native Method)
        at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
        at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
        at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
        at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
        at TopN.main(TopN.java:27)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
```

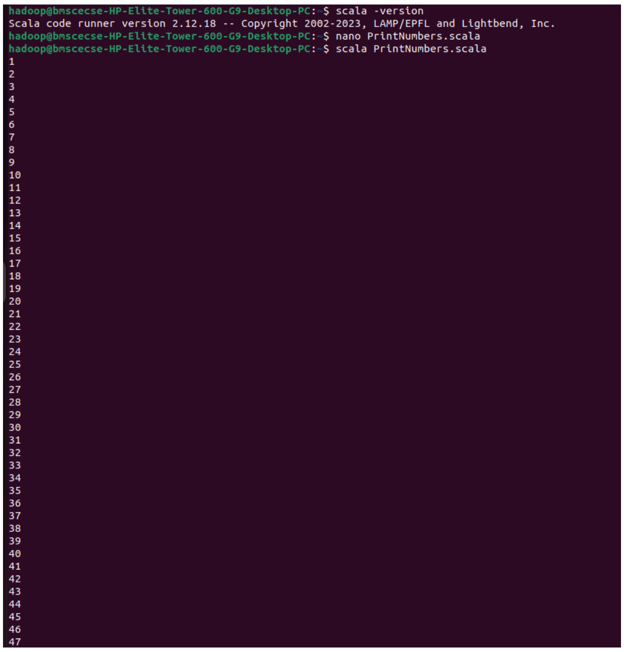
```
Note: utils/MiscUtils.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ jar -cvf TopN.jar *.class utils/*.class
added manifest
adding: TopN.class(in = 1388) (out= 764)(deflated 44%)
adding: TopNCombiner.class(in = 1587) (out= 662)(deflated 58%)
adding: TopNMapper.class(in = 1953) (out= 883)(deflated 54%)
adding: TopNReducer.class(in = 2532) (out= 1003)(deflated 60%)
adding: utils/MiscUtils$1.class(in = 944) (out= 494)(deflated 47%)
adding: utils/MiscUtils.class(in = 1231) (out= 693)(deflated 43%)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ nano ~/sample.txt
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -mkdir /topninput
mkdir: `/topninput': File exists
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -put ~/sample.txt /topninput
put: `/topninput/sample.txt': File exists
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -ls /topninput
Found 1 items
-rw-r--r--   1 hadoop supergroup         89 2026-04-24 14:58 /topninput/sample.txt
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop jar TopN.jar TopN /topninput/sample.txt /topnoutput
2026-04-24 15:04:22,185 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
Exception in thread "main" org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/topnoutput already exists
        at org.apache.hadoop.mapreduce.lib.output.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:164)
        at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:277)
        at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:143)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1678)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1675)
        at java.base/java.security.AccessController.doPrivileged(Native Method)
        at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
        at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
        at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
        at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
        at TopN.main(TopN.java:27)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
        at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
        at java.base/java.lang.reflect.Method.invoke(Method.java:566)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$ hadoop fs -cat /topnoutput/part-r-00000
how     5
your    4
is      4
brother 1
are     1
hi      1
sister  1
family  1
you     1
job     1
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/topn$
```

PROGRAM – 8

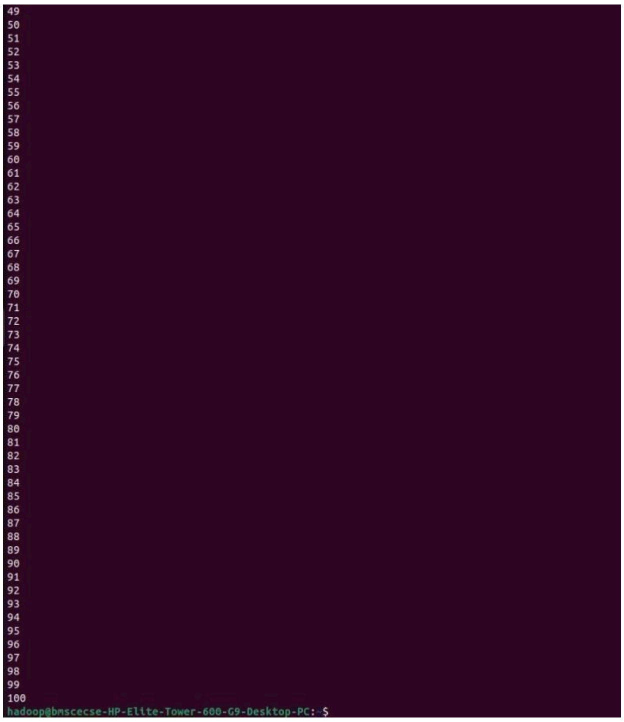
Write a Scala program to print numbers from 1 to 100 using for loop.

```
object ExampleForLoop1 {
    def main(args: Array[String]) { for(counter
        <- 1 to 100) {
            print(counter + " ")
        }
        println()
    }
}
```

Output:



```
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ scala -version
Scala code runner version 2.12.18 -- Copyright 2002-2023, LAMP/EPFL and Lightbend, Inc.
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ nano PrintNumbers.scala
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ scala PrintNumbers.scala
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
```



```
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$
```

PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

```
scala> val textFile = sc.textFile("D:\\Downloads\\spark-4.1.1-bin-hadoop3\\spark-4.1.1-bin-hadoop3\\spark-data\\wc.txt")
val textFile: org.apache.spark.rdd.RDD[String] = D:\Downloads\spark-4.1.1-bin-hadoop3\spark-4.1.1-bin-hadoop3\spark-data\wc.txt MapPartitionsRDD[1] at textF
ile at <console>:1

scala> val counts = textFile.flatMap(line => line.split(" ")).map(word => (word, 1)).reduceByKey(_ + _)
val counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:1

scala> val sorted = counts.collect().sortBy(-_._2)
val sorted: Array[(String, Int)] = Array((spark,6), (hello,4), (java,3), (python,2), (world,1))

scala> sorted.foreach(println)
(spark,6)
(hello,4)
(java,3)
(python,2)
(world,1)

scala> for((k,v) <- sorted) {
     |   if(v > 4) {
     |     print(k + " -> ")
     |     print(v)
     |     println()
     |   }
     | }
spark -> 6

scala> sorted.filter(_._2 > 4).foreach(println)
(spark,6)
```

PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

```
C:\Users\91988>spark-shell
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 4.1.1
      /_/

Using Scala version 2.13.17 (OpenJDK 64-Bit Server VM, Java 17.0.18)
Type in expressions to have them evaluated.
Type :help for more information.
Spark context Web UI available at http://LAPTOP-6DEICVVQ:4040
Spark context available as 'sc' (master = local[*], app id = local-1777009385296).
Spark session available as 'spark'.

scala> import spark.implicits._
import spark.implicits._

scala> spark.sparkContext.setLogLevel("ERROR")

scala> val lines = spark.readStream.format("socket").option("host", "localhost").option("port", 9999).load()
val lines: org.apache.spark.sql.DataFrame = [value: string]

scala> val words = lines.as[String]
val words: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val stopWords = Seq("is", "the", "a", "an", "and", "to")
val stopWords: Seq[String] = List(is, the, a, an, and, to)
```

```
scala> val cleaned = words.map { line =>
     |   val lower = line.toLowerCase
     |   val noSpace = lower.trim.replaceAll("\\s+", " ")
     |   val noPunct = noSpace.replaceAll("[^a-zA-Z ]", "")
     |   val filtered = noPunct.split(" ").filter(word => !stopWords.contains(word))
     |   val lemmatized = filtered.map {
     |     case w if w.endsWith("ing") => w.dropRight(3)
     |     case w if w.endsWith("ed")  => w.dropRight(2)
     |     case w => w
     |   }
     |   lemmatized.mkString(" ")
     | }
val cleaned: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val query = cleaned.writeStream.outputMode("append").format("console").option("checkpointLocation", "C:/tmp/check
point").start()
val query: org.apache.spark.sql.streaming.StreamingQuery = org.apache.spark.sql.execution.streaming.runtime.StreamingQueryWrapper@53f6a24

-------------------------------------------
Batch: 0
-------------------------------------------
+-----+
|value|
+-----+
+-----+

-------------------------------------------
Batch: 1
-------------------------------------------
+-----------------+
|            value|
+-----------------+
|she danc sing song|
+-----------------+
```

```
-------------------------------------------
Batch: 2
-------------------------------------------
+--------------------+
|               value|
+--------------------+
|cats are play in ...|
+--------------------+

-------------------------------------------
Batch: 3
-------------------------------------------
+--------------------+
|               value|
+--------------------+
|he walk talk teacher|
+--------------------+
```